Basics 11: Electric and Magnetic Fields

II - Electromagnetism foundations

Forces, potentials, flux, energy, and the field language used throughout plasma physics

Textbook draft, version 1.0. Electric and magnetic fields are the immediate agents that accelerate charged particles and transmit forces through a plasma. This chapter begins with Coulomb's law, develops electric potential and magnetic flux, and explains why field energy and field-line geometry are useful without treating field lines as material objects. This chapter is designed for a reader with no prior plasma-physics training. It distinguishes exact identities from physical approximations and connects the central equations to later simulation modules.

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1 Learning objectives

After completing this note, the reader should be able to:

- Distinguish force, field, potential, flux, and energy density.
- Derive the electric field from a scalar potential in electrostatics.
- Use the Lorentz force to interpret electric and magnetic effects.
- Compute electric and magnetic flux through simple surfaces.
- Check electromagnetic expressions using SI dimensions.
- Explain the limits of field-line pictures.

2 Prerequisites and reading strategy

- Basics 1-3.

On a first reading, emphasize physical meaning and dimensional checks. On a second reading, reproduce the derivations. Attempt the computational laboratory only after the limiting cases are understood.

3 From forces between particles to fields

For two point charges separated by $\mathbf{r} = \mathbf{x} - \mathbf{x}'$, Coulomb's law is

$$\mathbf{F}_{12} = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r^3}\mathbf{r}. \tag{3.1}$$

The electric field is defined as force per unit test charge:

$$\mathbf{E}(\mathbf{x}) = \lim_{q_t \to 0} \frac{\mathbf{F}}{q_t}. \tag{3.2}$$

The limit emphasizes that the test charge should not alter the source configuration. Superposition gives

$$\mathbf{E}(\mathbf{x}) = \frac{1}{4\pi\epsilon_0}\int \rho_q(\mathbf{x}')\frac{\mathbf{x}-\mathbf{x}'}{|\mathbf{x}-\mathbf{x}'|^3}\, d^3x'. \tag{3.3}$$

A field is therefore a rule assigning a vector to every point, not a stream of material moving along drawn lines.

4 Electric potential and electrostatic energy

In electrostatics, $\nabla \times \mathbf{E} = 0$, so the field can be written

$$\mathbf{E} = -\nabla\phi. \tag{4.1}$$

The potential difference is the work per unit charge required to move slowly against the field:

$$\phi(\mathbf{b}) - \phi(\mathbf{a}) = -\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\boldsymbol{\ell}. \tag{4.2}$$

Combining Gauss's law with $\mathbf{E} = -\nabla\phi$ gives Poisson's equation,

$$\nabla^2\phi = -\frac{\rho_q}{\epsilon_0}. \tag{4.3}$$

The electric energy density in vacuum is

$$u_E = \frac{\epsilon_0 E^2}{2}. \tag{4.4}$$

Only potential differences are physically observable in classical mechanics; adding a constant to ϕ changes no field.

5 Magnetic field and the Lorentz force

The force on a particle of charge q and velocity $\mathbf{v}$ is

$$\mathbf{F} = q\left(\mathbf{E} + \mathbf{v} \times \mathbf{B}\right). \tag{5.1}$$

Because $\mathbf{v} \cdot (\mathbf{v} \times \mathbf{B}) = 0$, a magnetic field alone does no instantaneous work:

$$\frac{d}{dt}\left(\frac{mv^2}{2}\right) = q\mathbf{v} \cdot \mathbf{E}. \tag{5.2}$$

Magnetic forces nevertheless redirect particles, change momentum transport, and allow a field to confine plasma. The magnetic energy density is

$$u_B = \frac{B^2}{2\mu_0}. \tag{5.3}$$

This energy density later appears as magnetic pressure, while spatial variation and curvature produce magnetic tension forces.

6 Flux, circulation, and field-line geometry

Electric and magnetic fluxes through a surface S are

$$\Phi_E = \int_S \mathbf{E} \cdot d\mathbf{A}, \qquad \Phi_B = \int_S \mathbf{B} \cdot d\mathbf{A}. \tag{6.1}$$

Flux depends on both field strength and surface orientation. Magnetic field lines are tangent to $\mathbf{B}$ and may be parameterized by

$$\frac{d\mathbf{x}}{d\ell} = \frac{\mathbf{B}}{B}. \tag{6.2}$$

Their visual density is conventionally used to indicate field magnitude, but this is a drawing convention. Since $\nabla \cdot \mathbf{B} = 0$, magnetic field lines do not begin or end at isolated magnetic charges in classical electromagnetism.

7 Electrostatic and inductive electric fields

When magnetic fields vary in time, the electric field need not be conservative:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \tag{7.1}$$

A useful decomposition is

$$\mathbf{E} = -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}, \qquad \mathbf{B} = \nabla \times \mathbf{A}. \tag{7.2}$$

The first term is potential-derived; the second is inductive. Plasma models often neglect one contribution under a stated ordering, but they should never assume $\mathbf{E} = -\nabla\phi$ without checking Faraday's law.

8 Scales relevant to plasma simulation

Typical comparisons include electric and magnetic force magnitudes,

$$\frac{|qE|}{|qvB|} \sim \frac{E}{vB}, \tag{8.1}$$

and field energy relative to thermal pressure,

$$\beta = \frac{p}{B^2/(2\mu_0)}. \tag{8.2}$$

Such ratios identify dominant physics. In numerical work, $\nabla\cdot\mathbf{B}$, energy balance, and unit consistency are primary diagnostics rather than cosmetic checks.

9 Computational laboratory

Implement Coulomb-field and potential calculations for a pair of charges. Verify numerically that $-\nabla\phi$ matches $\mathbf{E}$ away from the charges. Compute electric flux through nested spheres and magnetic flux through tilted planar surfaces. Compare finite-difference divergence errors under grid refinement.

10 Summary

- Fields convert source information into local force laws.
- Electrostatic electric fields derive from a scalar potential, while inductive fields do not.
- Magnetic fields redirect charged particles but do not directly change their kinetic energy.
- Flux and field-line descriptions are geometric tools whose limitations must be respected.
- Field energies and dimensionless ratios connect electromagnetism to plasma dynamics.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
$\mathbf{E}$	electric field	Force per unit positive charge.
$\mathbf{B}$	magnetic flux density	Field entering the magnetic Lorentz force.
ϕ	electric potential	Potential energy per unit charge.
ρ_q	charge density	Charge per volume.
Φ_E, Φ_B	field fluxes	Surface integrals of normal field components.
ϵ_0, μ_0	vacuum constants	Permittivity and permeability.

11 Exercises

1. Derive the electric field of a point charge from its potential.
2. Show explicitly that a magnetic force performs no work on a point particle.

3. Calculate the flux of a uniform magnetic field through a tilted circular loop.
4. Use dimensions to verify the units of u_E and u_B.
5. Explain why field lines can move in an ideal-MHD argument even though they are not material strings.

12 References

References

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